

ANARCHY

AI Platform

Roadmap + Competitor Analysis + Future Architecture

Technical Audit Report — File 8 of 8

3 yr	4	6	9.9
Roadmap	Competitors	Future Systems	Target Score

1. Competitor Analysis

Detailed comparison with Midjourney, Leonardo AI, VizMaker, and Krea AI

Anarchy AI operates in a competitive landscape dominated by general-purpose AI image tools and a small number of architecture-specific platforms. The key strategic insight is that Anarchy AI is the **ONLY** tool targeting the architecture workflow specifically with a native 3D software bridge. This is an undefended competitive position that must be reinforced before larger players copy it.

Feature	Midjourney	Leonardo AI	VizMaker	Krea AI	ANARCHY AI
Node-based canvas	✗	✗	✗	~ Partial	✓ Full
Architecture-specific AI	~ Generic	~ LoRA	✓ Arch focus	~ Generic	✓ Arch focus
3ds Max / Revit bridge	✗	✗	✗	✗	✓ Native
DXF vector export	✗	✗	✗	✗	✓ Built-in
Multiple AI models	✓ MJ only	✓ Many	~ Limited	✓ Many	✓ FLUX+NB2
Local AI (Ollama)	✗	✗	✗	✗	~ Planned
Desktop native app	✗	✗	✗	✗	✓ Electron
Offline capability	✗	✗	✗	✗	~ Planned
Generation history	✓ Full	✓ Full	✓ Full	✓ Full	~ Partial
Batch generation	✓ Full	✓ Full	✓ Full	✓ Full	✗ Missing
Real-time progress	✓ Full	✓ Full	✓ Full	✓ Full	✗ Fake timer
Collaboration	✓ Servers	✓ Teams	~ Basic	✓ Teams	✗ Missing
API access	✓ API	✓ API	✗	✓ API	✓ Replicate
Pricing (entry)	\$10/mo	Free tier	\$29/mo	Free tier	Credits
Iraqi market support	✗	✗	✗	✗	✓ ZainCash planned
Style presets	✓ Full	✓ Full	✓ Full	✓ Full	~ Basic
Client presentation	✗	~ Basic	~ Basic	~ Basic	✗ Missing
Plugin/extension system	✗	~ LoRA	✗	✗	~ Planned

Strategic Positioning

ANARCHY AI's undefended competitive moat: The 3ds Max / Revit native bridge + node-based canvas + DXF export pipeline is a combination that NO competitor has. This is the product's defensible advantage.

The positioning statement: 'VizMaker makes rendering faster. ANARCHY makes the architect smarter.' — This frames ANARCHY as the professional workflow tool, not just a faster generation button.

Risk: Midjourney and Leonardo AI have the resources to add architecture modes. The window to establish market presence is 12-18 months before this niche attracts major players.

2. Future Architecture Systems

Six architectural systems needed to reach 9.9/10 — design specifications

AI Job Queue System

The missing foundation of all AI operations

A priority-based, persistent job queue with retry, circuit breaker, and real-time status reporting. All AI operations (generate, upscale, variation) pass through this queue. Workers run in Electron main process. Frontend subscribes to job events via IPC.

- Bull Queue or p-queue (concurrency 3) in Electron main process
- Job schema: { id, type, payload, priority, status, retryCount, createdAt, updatedAt }
- Priority levels: manual > auto-variation > background
- Retry: 3 attempts with exponential backoff (1s, 2s, 4s)
- Circuit breaker: open after 5 failures, half-open after 30s
- Frontend: useJobQueue() hook subscribes to IPC job events
- UI: Job queue panel shows active, pending, completed jobs

Event Bus System

Decoupled communication between modules

A typed event bus that replaces direct function calls between canvas, services, and stores. This is the key architectural change that makes all components independently testable and allows features to be added without touching existing code.

- EventEmitter3 or Mitt as the event bus implementation
- Typed events: GenerationStarted, GenerationCompleted, NodeAdded, CanvasSaved, etc.
- Electron IPC bridge: main process events cross to renderer via ipcMain/ipcRenderer
- Components emit events; services subscribe; stores react
- Prevents circular dependencies: canvas → event bus → service (not canvas → service directly)
- Enables analytics: every event logged for debugging and metrics

Workflow Engine

Executable canvas workflows

The canvas becomes executable — not just a visual diagram, but a runnable pipeline. Click 'Run Workflow' and the engine traverses the node graph from source to output, executing each node's AI operation in dependency order.

- Topological sort of node graph to determine execution order
- WorkflowExecutor: traverse graph, execute each node sequentially or in parallel where possible
- Node interface: execute(inputs) => Promise — each node type implements this
- Progress tracking: emit progress events as each node completes

- Error handling: if a node fails, mark it red, stop downstream nodes, suggest retry
- Workflow templates: Save and share entire canvas workflows as .ana files

History Graph System

Non-linear generation history

Replace the flat history list with a visual history graph — a tree showing the branching evolution of images. Click any node to see how that image was created, which parent it came from, and what variations were explored.

- Store parent_id on every history record to build the tree
- History graph UI: tree visualization with branching paths
- Node detail: expand any history node to see prompt, model, seed, parent
- Re-branch: click any historical node to start a new variation branch
- Prune: collapse explored branches that didn't lead to good results
- Export: export any branch path as a PDF showing the creative evolution

Plugin Architecture

Extensible node type system

Allow third-party developers to add new node types to the canvas. A plugin defines its input/output schema, UI, and execute() function. Anarchy AI loads plugins at startup and they appear in the node palette.

- Plugin interface: { name, type, inputs, outputs, ui, execute }
- Plugin loader: Electron reads plugins from userData/plugins/ folder
- Plugin sandbox: plugins run in a restricted context (no filesystem access by default)
- Built-in plugins: migrate all 10 GhostNode types to plugin format first
- Plugin API: access to canvas state, history, and Replicate API via provided hooks
- Future: plugin marketplace with signed, reviewed plugins

Canvas Engine 2.0

WebGL-accelerated canvas for 500+ nodes

Replace the HTML5 Canvas LiteGraph renderer with a WebGL-based engine for large-scale canvases. This enables smooth 60fps rendering with 500+ nodes, GPU-accelerated image display in nodes, and support for real-time video frames in the future.

- Pixi.js or Three.js for WebGL canvas rendering layer
- Migrate LiteGraph node definitions to WebGL sprite-based rendering
- GPU texture management: images stored as WebGL textures (not DOM elements)
- Level-of-Detail: simplified rendering at zoom < 30% for smooth pan/zoom
- Viewport culling: GPU clip-space test replaces CPU bounding box check
- Target: 60fps on 200 nodes; 30fps on 500 nodes; 15fps on 1000 nodes

3. One-Year Roadmap

Month-by-month milestones to reach 9.0/10 and public launch

Period	Focus	Key Work	Score	Milestone
Month 1-2	Stabilization	Fix all CRITICAL bugs, complete LiteGraph migration, add Error Boundaries, implement AI Job Queue	7.5/10	Private alpha release to 10 trusted architects
Month 3-4	Architecture	Break God Components, add granular Zustand selectors, performance sprint, code splitting	8.0/10	Closed beta — 50 users, credit-based pricing live
Month 5-6	UX Overhaul	Node palette, workflow templates, history pagination, onboarding tour, real progress indicators	8.5/10	Public beta launch — waitlist open
Month 7-8	Security	Complete RLS, code signing, MFA, webhook verification, rate limiting, security audit	8.8/10	Production v1.0 release
Month 9-10	Features	Revit bridge, local AI (Ollama), batch generation, library collections, Google OAuth	9.0/10	v1.1 — Professional tier launch
Month 11-12	Scale	Event Bus, Workflow Engine, History Graph, plugin architecture skeleton, Sentry monitoring	9.3/10	v1.5 — Power User release

4. Three-Year Vision

Strategic milestones to reach 9.9/10 and category leadership

Year	Theme	Key Capabilities	Score	Target
Year 1	Foundation	Stable, secure, production-ready platform. Node canvas, 3ds Max bridge, AI queue, 3 AI models.	9.3/10	1,000 active architects globally
Year 2	Ecosystem	Plugin marketplace, Workflow Engine, History Graph, Canvas Engine 2.0, real-time collaboration, Revit + SketchUp bridges.	9.7/10	10,000 active users; \$1M ARR
Year 3	Platform	ANARCHY becomes the operating system for architectural visualization. Third-party plugins, firm licensing, white-label API for AEC software vendors.	9.9/10	100,000 users; category leader

5. How Anarchy AI Reaches 9.9 / 10

The complete formula — architecture + product + market

9.9/10 is not a technical score — it is a product-market-architecture score. The technical foundation enables the product vision; the product vision earns the market position. The three pillars must advance together.

Technical Excellence (Architecture) Current: 6.8 → Target: 9.5	Complete LiteGraph migration. God Components split. AI Job Queue. Event Bus. Test coverage 80%. Performance: 60fps on 200 nodes. Zero production-critical bugs.
Product Completeness (Features) Current: 6.4 → Target: 9.5	Real-time progress. Cancel. Batch generation. Workflow templates. History graph. Library collections. Plugin architecture. Revit/SketchUp bridges. Collaboration.
Market Position (UX + Distribution) Current: 6.8 → Target: 9.8	Professional UX matching Figma quality bar. Active community of 10,000+ architects. VizMaker users migrating to ANARCHY. Cited in architecture schools and firms.

Final Statement

Anarchy AI has everything needed to win in the architectural AI visualization market. The product concept is differentiated (node-based + 3ds Max bridge). The visual identity is strong. The technical stack is appropriate. The competitive window is open. What stands between the current state and 9.9/10 is not a gap in vision — it is the execution discipline of fixing 50 known issues in 12 focused sprints. The path is clear. The work is defined. The market is waiting.

End of Audit Series — File 8 / 8

This concludes the complete Anarchy AI Technical Audit Series — 8 files, ~140 pages. Files covered: Architecture, Services/Stores/Components, Functional, Performance, Security, UX, Top 50 Problems + Top 100 Improvements, and Roadmap. Total issues documented: 50 problems, 100 improvements, 15 security vulnerabilities, 10 expected bugs, 20 performance issues, 40 UX improvements.